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Unit 5 : Session 3 : SLO 1 | Implementing Automata with the Python automata Library

I. Questions & Answers

1. Library is the automata-lib library (pip install automata-lib), specifically automata.fa.dfa.DFA and automata.fa.nfa.NFA, is used to define DFAs/NFAs in Python.
2. Acceptance check is dfa.accepts_input("0101") returns True/False; dfa.read_input("0101") returns the final state.
3. Defining a transition is transitions are given as a nested dictionary: transitions={"q0": {"0": "q1", "1": "q0"}, "q1": {"0": "q1", "1": "q0"}} i.e. $\hat{T}(\text{state}, \text{symbol}) = \text{next state}$.

II. Example

- from automata.fa.dfa import DFA
- dfa = DFA(states={"q0", "q1"}, input_symbols={"0", "1"},
- transitions={"q0": {"0": "q1", "1": "q0"}, "q1": {"0": "q1", "1": "q0"}},
- initial_state="q0", final_states={"q1"})
- print(dfa.accepts_input("1010")) # True is string ends with 0